MandiMitra: An Explainable Framework for Evaluating Short-Horizon Agricultural Market Price Forecasts Using Historical Indian Mandi Data

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Abstract

Price-forecasting tools for Indian farmers and traders are increasingly promoted on the strength of machine-learning accuracy, but the evaluations behind such claims are often hard to audit. We present MandiMitra, an open-source framework and bilingual (English/Hindi) dashboard that evaluates 1–3-day-ahead forecasts of daily wholesale onion and tomato prices from the Department of Consumer Affairs series for Jaipur, 2009–2015, against the simplest benchmark, “tomorrow equals today”. Five candidates — Naive, ARIMA, Random Forest, Gradient Boosting and a Random Forest that predicts the price change — are selected by rolling-origin cross-validation inside the development period and scored once on a held-out final period (February 2014 – April 2015, about 200 days per horizon). Features use only information available at the forecast origin, targets are scored only on days the market actually reported, and training rows whose target dates cross a boundary are removed. The model chosen in advance never had a lower test RMSE than Naive (Onion: Naive was chosen at all three horizons; Tomato: −5.66%, −0.53% and −0.31%), and Naive had the lowest test MAE at all six pair-horizons. Diebold–Mariano tests with Holm adjustment found no comparison, out of 48, in which a model was significantly more accurate than Naive. A first version of the project, which picked the lowest-test-error model, had reported a 7.33% improvement; that difference (Diebold–Mariano $p = 0.505$) is not distinguishable from noise. On 83–85% of reported days the quoted price equalled the previous report. Empirical 80% forecast ranges covered 55–74% of test outcomes. We argue that, for short-horizon mandi prices from price history alone, an honest benchmark comparison and transparent uncertainty reporting are more useful to users than a model leaderboard, and we release every artifact needed to reproduce the results.

Keywords: agricultural price forecasting; naive benchmark; time-series cross-validation; data leakage; Diebold–Mariano test; India; onion; tomato; explainable decision support.

1. Introduction

Onion and tomato prices in India are volatile. In the series studied here, Jaipur’s wholesale onion price rose from Rs. 600–900 per quintal in June 2013 to Rs. 4,500 on 21 October 2013 and fell to about Rs. 800 by December. Such movements matter to farmers deciding when to sell and to traders deciding how much to hold, and they motivate a growing number of price-prediction tools.

Accuracy claims for such tools are only as good as the evaluation behind them. Three problems recur in applied forecasting. First, a model may be chosen because it had the lowest error on the same data used to report its accuracy, which overstates performance. Second, information from after the forecast origin can leak into features, targets or preprocessing (Kaufman et al., 2012). Third, results are often reported without the simplest benchmark: for many economic series, “the next value equals the current one” is hard to beat (Meese and Rogoff, 1983), and a comparison of statistical and machine-learning methods on the M3 data found the machine-learning methods dominated on accuracy across horizons (Makridakis, Spiliotis and Assimakopoulos, 2018).

This paper describes MandiMitra, a small, fully reproducible framework built to make those three problems visible and testable for Indian mandi prices, and reports what it finds on the data that passed its quality gate. The contribution is not a new forecasting model. It is (i) an evaluation protocol whose decision rules were written before the corresponding results were seen and whose researcher degrees of freedom are disclosed; (ii) automated tests that would fail if leakage, test-set selection or mismatched scoring rows were reintroduced; (iii) a worked example in which correcting the protocol removed a reported improvement; and (iv) a bilingual dashboard that presents these results, their uncertainty and the data’s limitations to non-specialists.

2. Problem statement and research questions

Given a daily wholesale price series for one commodity in one mandi, with missing days, we ask whether forecasts made at the end of day t for days $t+1$, $t+2$ and $t+3$ can be made more accurate than repeating the price reported at t, using only the price history.

- **RQ1.** Do more complex models (ARIMA, tree ensembles) improve on the naive baseline for the available series, when the model is chosen before the final test period is scored?
- **RQ2.** How does relative performance change with horizon (1–3 days)?
- **RQ3.** How do missing reports and stale quotes affect what can be learned and how results should be presented?
- **RQ4.** Can forecast uncertainty be communicated reliably with a simple distribution-free method on this data?

We do not ask whether the forecasts improve farmers’ incomes; no user study was conducted (Section 10).

3. Related work

Benchmarks and accuracy measures. Hyndman and Athanasopoulos (2021, §5.2) define the naive forecast as the last observation and the seasonal naive forecast as the last observation from the same season; both are standard benchmarks. Hyndman and Koehler (2006) review the properties and pitfalls of scale-dependent and percentage error measures such as MAE, RMSE and MAPE. Meese and Rogoff (1983) is the classic demonstration, for exchange rates, that structural models did not beat a random walk out of sample. In the M4 competition (Makridakis, Spiliotis and Assimakopoulos, 2020), simple benchmarks were included as reference points for 61 methods on 100,000 series; we use it only as context.

Makridakis, Spiliotis and Assimakopoulos (2018) compared machine-learning and statistical methods on 1,045 monthly M3 series and found the machine-learning methods were dominated by the statistical ones on both accuracy measures and at all horizons.

Evaluating time-series forecasts. Tashman (2000) reviews out-of-sample evaluation, including rolling-origin designs. Bergmeir and Benítez (2012) study cross-validation choices for time-series predictors. Cerqueira, Torgo and Mozetič (2020) compare performance-estimation methods empirically and find that for non-stationary series, out-of-sample (holdout-style) estimates were the most accurate. Kaufman et al. (2012) formalise leakage and how to detect and avoid it. Diebold and Mariano (1995) propose a test of equal predictive accuracy, and Harvey, Leybourne and Newbold (1997) give a small-sample correction, which we use. Lei et al. (2018) describe split-conformal predictive inference; its coverage guarantee rests on exchangeability, which a non-stationary price series does not satisfy.

Agricultural prices in India. Birthal, Negi and Joshi (2019) associate onion price volatility in India with uncertainty in market arrivals rather than production shocks, and find that export restrictions had no immediate cooling effect — a reminder that price history alone omits the main drivers. Paul et al. (2022) compare machine-learning methods (GRNN, SVR, Random Forest, Gradient Boosting) with ARIMA for brinjal prices in 17 Odisha markets, find GRNN best, and use Diebold–Mariano tests; the abstract does not report a random-walk benchmark. Avinash et al. (2024) forecast weekly tomato, onion and potato prices for Azadpur (Delhi), 2006–2023, with hidden-Markov-model/deep-learning hybrids against standalone deep models; no naive benchmark is reported and Diebold–Mariano significance holds only in some cases. Our work differs in scope (daily, 1–3 days, two series) and purpose: it asks first whether any candidate beats the naive benchmark under a protocol that cannot select on the test period, and reports significance for every comparison.

Tools. We use scikit-learn (Pedregosa et al., 2011) for Random Forest (Breiman, 2001) and Gradient Boosting (Friedman, 2001), and statsmodels (Seabold and Perktold, 2010) for ARIMA.

4. Data, provenance and licensing

Source. Daily wholesale prices of onion and tomato published by the Department of Consumer Affairs, Price Monitoring Cell, on the Open Government Data Platform India (data.gov.in), 1 January 2009 to 28 April 2015, for 73 centres nationally (DCA, 2015). Unit: Rs. per quintal, as published. A price of 0 is the source's code for "not reported".

Provenance. The two official files are stored unaltered with SHA-256 hashes. A script extracts the Rajasthan centres; a test proves that, restricted to Jaipur/Onion, it reproduces the project's original raw file byte for byte. Every input and output file is hashed in a generated run manifest.

Licensing. The platform's Terms of Use require that the source be prominently acknowledged and refer to per-dataset licence metadata; the catalogue records for these datasets carry no explicit licence field. The Government Open Data License – India (GODL, 2017) is the licence associated with the platform; its text could not be read from the official

page during this study (Section 10). The code is MIT-licensed; that licence does not cover the data.

Series that passed the gate. A series is evaluated only if it has at least 400 rows with a complete feature vector and a genuinely reported 1-day target — a threshold fixed before any Rajasthan series was examined.

Table 1. Coverage of the Rajasthan series (1 Jan 2009 – 28 Apr 2015).

Series	Raw rows	Reported prices	Reporting %	Usable 1-day rows	Gate
Onion / Jaipur	1,551	1,288	55.8	1,002	evaluated
Tomato / Jaipur	1,551	1,228	53.2	995	evaluated
Onion / Jodhpur	1,122	625	37.4	315	excluded
Tomato / Jodhpur	1,122	626	37.4	322	excluded
Onion / Kota	1,105	470	45.8	241	excluded
Tomato / Kota	1,105	470	45.8	245	excluded

Reporting % is days with a reported price over calendar days in the series' own span. Source: `docs/DATA_REPORT.md`, generated from the files.

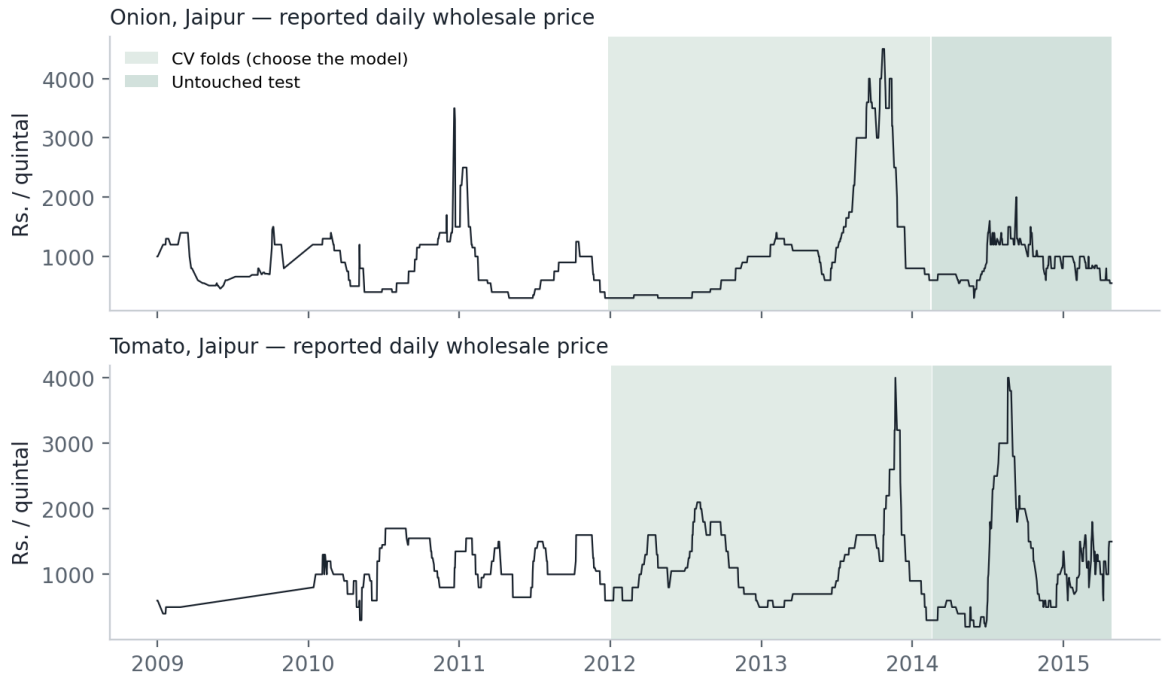


Figure 1. Reported daily wholesale prices, Jaipur, with the cross-validation span and the untouched test period (1-day horizon boundaries).

5. Cleaning and feature engineering

Zeros are treated as missing; prices outside Rs. 50–100,000 per quintal are rejected; duplicates are removed. Each series is placed on a continuous daily calendar so that a one-day lag always refers to the previous calendar day — an earlier version of the project dropped gap rows before computing lags, which silently made "yesterday" mean "the previous report"; a calendar-lag check now guards against this. A missing day may carry the last reported price forward as an *input* for at most two days (chosen by measuring usable rows: one day gave 391 usable 1-day rows on Onion/Jaipur, two days 1,002, three days 1,145); longer gaps stay empty. About 35% of feature rows use a carried-forward price.

The decision point is the end of day t. Twelve features use data up to and including t: the price at t; lags of 1, 3 and 7 days; 3- and 7-day moving averages; 7-day volatility; the 1-day return; deviation from the 7-day average; days since the last report; day of week; month. Windows measured in calendar days require at least 60% of their days to be reported. Targets are the prices at $t+1$, $t+2$, $t+3$, and a row is used for training or scoring **only if the target day was actually reported**. Before this rule, about 40% of 1-day targets would have been carried-forward values, which would have handed the naive benchmark free exact hits.

Leakage defences, each enforced by an automated test: backward-only windows (a code scan fails the build if back-filling or interpolation appears); direct multi-horizon models (no prediction is fed back as an input); a truncate-and-compare proof that features before a cut date are unchanged when later data are deleted; a perturbation test in which multiplying future prices by ten leaves earlier features unchanged; and a causal ARIMA filter whose forecasts are shown not to depend on later prices.

6. Forecasting methods

- **Naive:** $\widehat{y}_{t+h} = y_t$.
- **ARIMA:** order selected by AIC on the fitting window from seven candidates (all with one difference); h-step forecasts from every origin by a single causal Kalman-filter pass with parameters estimated on the fitting window.
- **Random Forest** (300 trees, depth 6, minimum leaf 8, square-root feature sampling) and **Gradient Boosting** (200 stages, learning rate 0.05, depth 3, minimum leaf 8, subsample 0.9), one model per horizon, predicting the price level.
- **Random Forest-delta:** as Random Forest but predicting $y_{t+h} - y_t$, then adding y_t back. Level-predicting trees output averages of training targets and so cannot forecast above the highest price they were trained on; in the cross-validation fold containing the October 2013 spike, the level trees could not predict above Rs. 2,197 per quintal.

The fifth candidate was admitted by an experiment run under a rule written before it ran: a change-predicting variant joins the candidates only if its mean cross-validation RMSE beats Naive at one or more horizons. Random Forest-delta did so on Tomato/Jaipur (1 and 3 days); a Gradient Boosting variant did not anywhere and was not added.

7. Evaluation protocol

The last 20% of distinct dates of each series form the **final test period** (17 February 2014 – 27 April 2015 for Onion at 1 day; boundaries differ slightly by series and horizon), scored once after selection. Inside the earlier development period, **rolling-origin cross-validation** (Tashman, 2000) uses four expanding folds over its second half: each fold fits on everything before its origin and is scored on the following block. **The selection rule, fixed before the results were seen, is the lowest mean per-fold RMSE.** A mean of folds, rather than a pooled RMSE, prevents one spike fold from deciding alone. An **embargo** removes any training row whose target date reaches past a fold origin or the test cut-off. Within each fold and on test, every candidate is scored on **identical rows**. Final models are refitted on the whole development period.

Point accuracy is reported as MAE and RMSE in Rs. per quintal; MAPE is in the outputs but is not used for decisions because it is asymmetric and unstable near zero (Hyndman and Koehler, 2006). R^2 is not reported on screen because on a series that mostly repeats itself it is high for any model.

Significance is assessed with the Diebold–Mariano test (Diebold and Mariano, 1995) with the Harvey–Leybourne–Newbold correction (Harvey, Leybourne and Newbold, 1997), autocovariances to lag $h-1$, on the identical test days, under squared and absolute loss, with Holm adjustment across the four model-versus- Naive comparisons in each pair-horizon.

For uncertainty, an 80% range is formed from the selected method's out-of-fold cross-validation errors (a split-conformal-style construction, Lei et al., 2018) and checked once on the test period. A range "holds up" if test coverage is within ten percentage points of nominal.

Development history and disclosed degrees of freedom. The first version of the project (September 2026) reported the lowest-test-RMSE model as "best" and split on the feature date without an embargo; its results are preserved, not overwritten. The current protocol replaced it. Three further judgement calls were made with knowledge of non-test results and are disclosed: (a) the validation design was changed from a single block to rolling folds after the single block was seen to fall on one price spike; (b) the fifth candidate was admitted on cross-validation; (c) the tree settings were fixed under the first protocol, when their effect on the old test window had been observed. Item (c) is audited in Section 8.4.

8. Results

8.1 Main comparison (RQ1, RQ2)

Table 2. Final test period. Selection by cross-validation; test scored once, afterwards. Rs. per quintal.

Series	h	Chosen by CV	CV mean fold RMSE	Test RMSE	Naive test RMSE	vs Naive	Test MAE	Naive test MAE	n
Onion/Jaipur	1	Naive	61.53	94.27	94.27	0.00%	39.75	39.75	200
Onion/Jaipur	2	Naive	90.40	110.29	110.29	0.00%	53.23	53.23	201
Onion/Jaipur	3	Naive	122.77	127.93	127.93	0.00%	65.85	65.85	205
Tomato/Jaipur	1	RF-delta	75.68	127.33	120.51	−5.66%	70.92	54.27	199
Tomato/Jaipur	2	ARIMA	106.03	179.10	178.16	−0.53%	99.25	90.55	201
Tomato/Jaipur	3	ARIMA	132.99	227.38	226.68	−0.31%	131.68	123.77	204

Source: `outputs/best_model.csv, outputs/model_comparison.csv.` *"vs Naive" is the percentage reduction in test RMSE; negative means worse than Naive.*

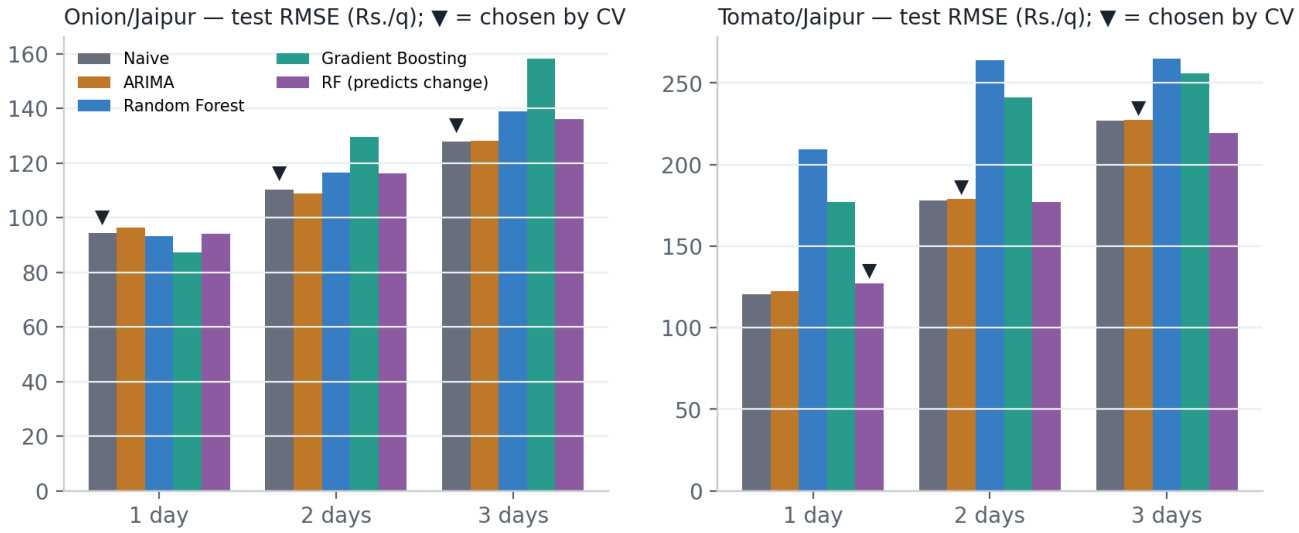


Figure 2. Test RMSE of every candidate; ▼ marks the model chosen by cross-validation.

The model chosen in advance never had a lower test RMSE than Naive (RQ1). Naive had the lowest test MAE at all six pair-horizons. Errors grow with horizon for every method (RQ2), and no horizon shows a pattern in which a complex model becomes useful.

Hindsight winners. Looking at the test period afterwards, a different model had the lowest test RMSE at five of six pair-horizons: Gradient Boosting for Onion at 1 day (87.26), ARIMA for Onion at 2 days (108.94), Naive for Tomato at 1 day (120.51), and Random Forest-delta for Tomato at 2 and 3 days (176.91, 219.18). These are reported for transparency and were not used for selection. Gradient Boosting's cross-validation RMSE for Onion at 1 day was 229.03 — almost four times Naive's 61.53 and second-worst of the five (only Random Forest, 275.19, was worse) — illustrating how unstable such a "winner" is.

8.2 Significance

Table 3. Diebold–Mariano tests, every model against Naive on the identical test days, Holm-adjusted within each pair-horizon (24 comparisons per loss).

Loss	Model significantly better (p<0.05)	Naive significantly better	Not distinguishable
Squared error	0	6	18
Absolute error	0	24	0

Source: `outputs/experiments/dm_test.csv`.

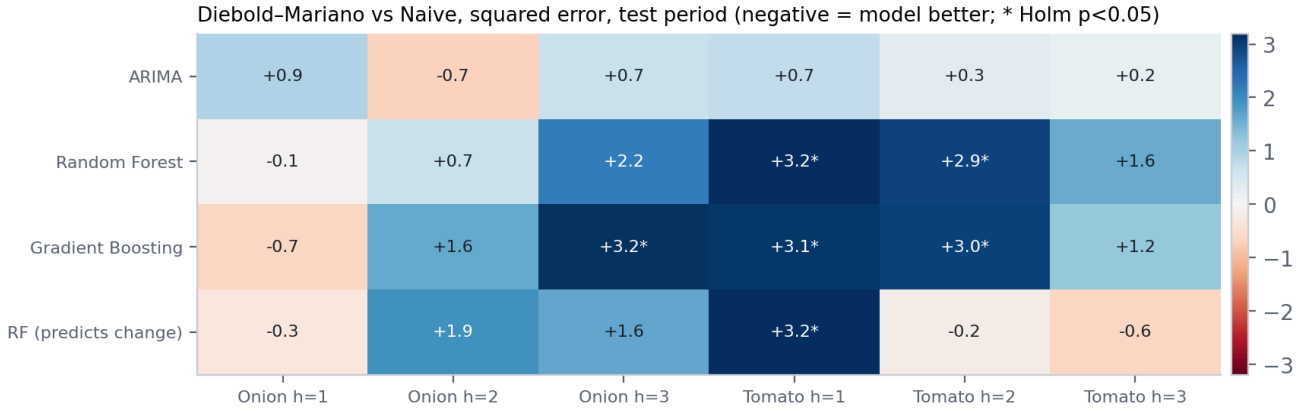


Figure 3. Diebold–Mariano statistics (squared error).

No comparison, under either loss, found a model significantly more accurate than Naive. The Tomato 1-day choice (Random Forest-delta) was significantly worse than Naive under squared error (DM = 3.18, Holm-adjusted p = 0.007). The first version's headline — Gradient Boosting 7.33% better than Naive for Onion at 1 day — corresponds to DM = −0.67, p = 0.505. The absolute-loss result partly reflects that Naive is exactly right on every day the quoted price does not change.

8.3 A second baseline

A weekly seasonal naive forecast ($\widehat{y}_{t+h} = y_{t+h-7}$) was evaluated on the same folds under a rule fixed beforehand (it would replace Naive as the reference only if better at every horizon of a series). Its mean cross-validation RMSE was far higher: Onion 169.5 / 160.5 / 168.2 against Naive's 63.0 / 84.9 / 117.3, Tomato 215.0 / 212.9 / 213.3 against 79.4 / 109.7 / 136.5 (h = 1, 2, 3; rows where both baselines exist). Naive remains the reference.

8.4 Sensitivity to the tree settings

Re-running every tree candidate with scikit-learn's default settings on the same folds changed one selection: Tomato at 1 day would choose ARIMA instead of Random Forest-delta. ARIMA's test RMSE there (122.16) is also above Naive's (120.51), so no conclusion changes. Random Forest-delta's admission does depend on the regularised settings (its mean CV RMSE at Tomato h = 1 is 91.2 with defaults, against Naive's 76.9).

8.5 Data quality (RQ3)

On 84.8% (Onion) and 82.8% (Tomato) of reported Jaipur days, the quoted price equalled the previous report (Figure 5). Prices move in occasional steps rather than continuously,

which favours the naive benchmark on ordinary days and leaves the large steps — the costly errors — unpredictable from price history. About 44–47% of calendar days have no report, so 35% of model inputs use a carried-forward price; none is ever used as a target.

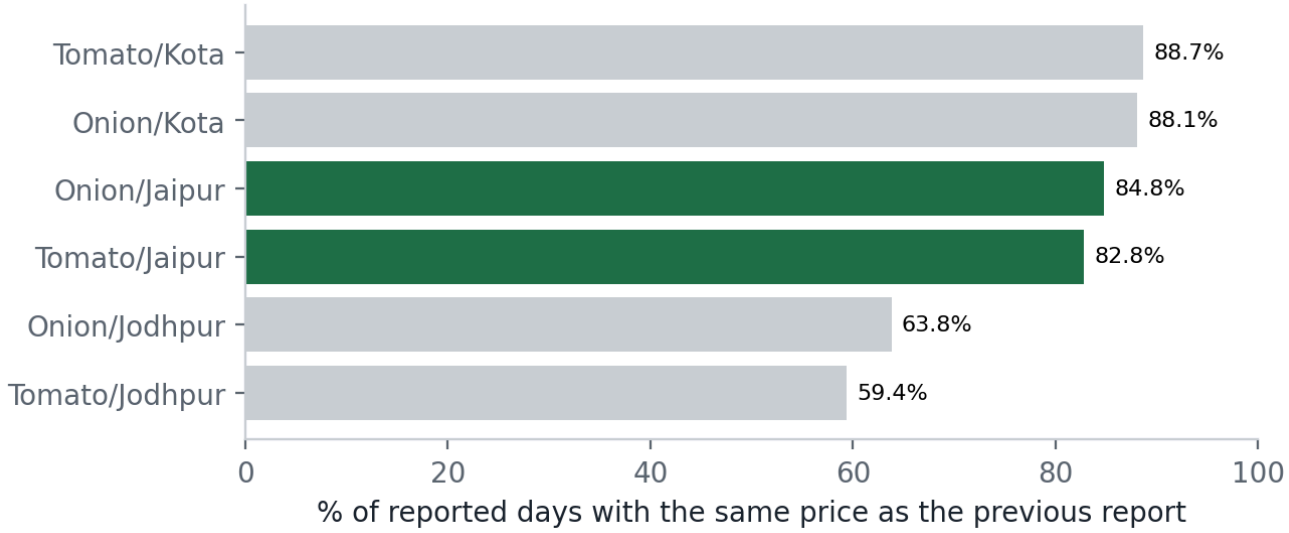


Figure 5. Share of reported days with the same price as the previous report.

8.6 Uncertainty (RQ4)

Empirical 80% ranges built from cross-validation errors covered 74.0%, 64.7% and 66.8% of Onion test outcomes and 55.3%, 56.7% and 69.6% of Tomato outcomes (h = 1, 2, 3). None held up by the pre-set criterion, and the range nearest to nominal (Onion, 1 day) had zero width because most cross-validation errors were exactly zero. The dashboard therefore reports the coverage check instead of drawing a band.

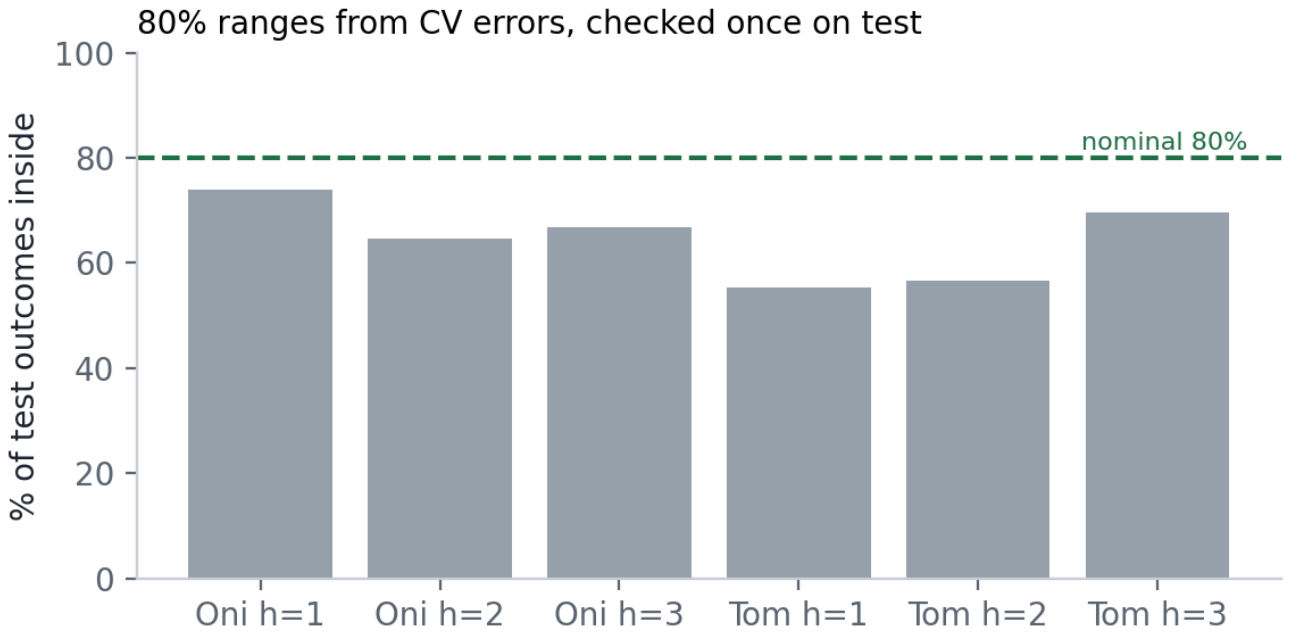


Figure 4. Coverage of the 80% ranges on the test period.

9. Discussion

For these two series, price history alone did not support forecasts better than “tomorrow equals today” at 1–3 days. This agrees with the long-standing difficulty of beating random-walk-type benchmarks for economic prices (Meese and Rogoff, 1983) and with large-scale comparisons in which machine-learning methods did not outperform statistical ones (Makridakis et al., 2018). It also fits the finding that onion price volatility is associated with arrivals uncertainty (Birthal et al., 2019): the information that might predict a price step is not in the price series.

The corrected protocol changed a reported 7.33% improvement into no detectable improvement. Nothing about the data or models changed; only the rule for choosing the winner did. This is the practical case for selecting on validation data, scoring the test once, testing significance, and publishing the hindsight winners alongside the chosen model rather than in its place.

For users, the useful outputs are therefore not a confident forecast but: which simple reference to trust; how often the price actually moves; how uncertain any forecast is; and how stale the data are. The dashboard is built around those, in English and Hindi, with every figure traceable to a file.

10. Limitations and threats to validity

- **One mandi, two commodities, 2009–2015.** Results may not hold elsewhere or now. They are not evidence that forecasting is impossible with richer inputs.
- **Retrospective only.** No forecast was checked prospectively.
- **Price history only.** No arrivals, weather, policy or festival data.
- **Gaps and stale quotes.** Reporting density near 55%; most reported prices unchanged. Part of this may reflect recording practice rather than trading.
- **Test window length.** About 200 scored days per horizon limits power; “not distinguishable” is not “equal”.
- **Regime change.** The test period follows the 2013 spike and is more volatile than most of the development period.
- **Researcher degrees of freedom.** Disclosed in Section 7 and audited where possible (Section 8.4); a prospective study would remove them.
- **Licensing.** The licence text was not read from the official page.

11. Responsible use and societal impact

The dashboard states on every screen that the data are historical and that nothing it shows is financial, trading or farming advice. Forecasts are shown next to the last reported price and the naive reference, never alone. No personal data are collected; the app has no accounts or tracking. A tool that overstated accuracy could lead a farmer to hold or sell at the wrong time; the design choice here is to understate rather than overstate. Any future deployment with current data should be preceded by a consented user study (the project includes a plan) and should keep the benchmark comparison visible.

12. Reproducibility

All code, the two untouched source files, every derived file and every result are in the public repository, with pinned dependencies, a fixed seed and a generated manifest of SHA-256 hashes. Rebuilding from the government files gives byte-identical results under CPython 3.11 and 3.12 on Linux; on other platforms only ARIMA output moves, by hundredths of a rupee, within a tolerance checked by a script. 154 automated tests cover leakage, chronology, embargo, identical scoring rows, provenance, the manifest, data validation and the dashboard. Continuous integration rebuilds everything from the source files on each push.

```
python scripts/extract_rajasthan.py
python scripts/build_features.py
python scripts/run_pipeline.py --all-pairs
python scripts/experiment_significance.py
python -m pytest
```

13. Conclusion and future work

Under a protocol that cannot select on the test period, none of four non-trivial forecasting methods beat “tomorrow equals today” for daily onion and tomato wholesale prices in Jaipur at 1–3 days, and no difference in a model’s favour was statistically detectable. We suggest that applied agricultural price-forecasting work report a naive benchmark, select without the test period, test significance, and publish its hindsight winners.

Future work: (1) current data — the framework includes a validated client for the official AGMARKNET-sourced daily price API, pending an access key; (2) more markets and commodities through the same gate; (3) arrivals and weather inputs, where an edge is more plausible; (4) prospective evaluation, scoring forecasts against prices published after they were made; (5) a consented user study of whether the explanations help traders and farmers.

Acknowledgements and declarations

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